Camping and Wilderness Survival: Combined Text Summary

Condensed from four guides: US Army FM 21-76 Survival (2006), You Alone in the Maine Woods (11th ed.), Wilderness Survival (Survivor Library), and Ultimate Survival Skills Guide (Joe Marshall).

1. Mindset and priorities

SURVIVAL acronym (FM 21-76)

- **S**ize up the situation (surroundings, physical condition, equipment).
- **U**se all your senses; undue haste makes waste.
- **R**emember where you are (keep track of landmarks and your position).
- **V**anquish fear and panic.
- **I**mprovise with what you have.
- **V**alue living; keep the will to survive.
- **A**ct like the natives; observe how local people and animals cope.
- **L**ive by your wits, but for now learn basic skills.

Order of priority

Treat life-threatening injury first, then protect yourself from the elements (shelter, clothing, fire), then signal for rescue, then water, then food. Rules of thumb: about 3 minutes without air, 3 days without water, about 3 weeks without food (a person can survive on very little food for a long time; Maine Woods and Survivor Library both stress that a week or more without food is not an emergency, while thirst and cold are).

Survivor Library's seven enemies

- Fear, pain, cold, thirst, hunger, fatigue, boredom and loneliness. Name them, expect them, and plan for each.

Mental control

- Stress is normal. Stay busy with useful tasks; set small goals (build shelter, gather wood, signal).
- If you panic, stop and sit down. Use **STOP**: Stop, Think, Observe, Plan.
- Keep morale up with routine, rest and warmth. Avoid dwelling on discomfort.

2. Planning and kit

Before you go

- Tell someone: what you are doing, where (route and trailhead), who is with you, your vehicle description, and when you will be back. Do not hide the vehicle.
- Check the forecast, carry a map and compass and know how to use them, and match the trip to your fitness.

- Carry the essentials even on short outings.

On your person at all times

- Knife, map, compass, waterproof fire kit (matches in a waterproof case, lighter, firestarter), whistle.
- Optional on-person items: small flashlight, signal mirror, water purification tablets, space blanket.

In your pack (additional items)

- Extra layers and a hat, rain gear, sturdy bags (contractor bags are excellent emergency sleeping bags or ponchos), headlamp with spare batteries, cordage, duct tape.
- First aid kit, water bottle(s) and purification method, high-energy food, sunscreen and insect repellent.
- Emergency blanket or foil, a small saw or folding saw, tarp, spare socks, toilet supplies.
- Survivor Library: keep total pack weight around 18 kg (40 lb) at most; use a checklist and pack the same way every time.

Survival tin (FM 21-76)

Use a tough, waterproof, corrosion-resistant case that can also serve as a cup. Include: water purification tablets, matches and striker, candle, small compass, fishing line and hooks, wire saw, needle and thread, safety pins, razor blade or scalpel, button compass, bandages and antiseptic, signaling mirror, whistle. Keep it small enough to carry in a pocket.

72-hour / bug-out bag (Marshall)

- Water and a way to purify it, 3 days of non-perishable food, first aid kit, flashlight and spare batteries, radio, multi-tool, warm clothing and rain gear, lighter and matches, important documents, cash, personal medications.
- Rule of thumb: store at least 3 days of supplies for an emergency, ideally 3 weeks for larger disasters.

3. Clothing and weather

Layering

- Use three layers: a wicking base layer, an insulating mid layer (fleece or wool), and a wind/waterproof shell. Add or remove layers to avoid sweating.
- Avoid cotton: it holds moisture and loses insulation. Wool and synthetics keep some warmth when wet. Down loses almost all warmth when wet.
- COLD (FM 21-76): keep Clean, avoid Overheating, wear Loose layers, keep Dry.
- The head can lose a large share of body heat (the manual says 40 to 45 percent; Maine Woods says 25 to 70 percent depending on conditions). Always carry a warm hat; cover the neck and face in wind.
- Footwear: boots should fit with room for a thick sock, be broken in before the trip, and be waterproof or treated. Carry spare dry socks.

- Wind chill matters: for example, minus 10 C with a 28 km/h wind feels about minus 23 C.

Weather awareness

- Watch the sky: rapidly building dark clouds, a sudden temperature drop or a change in wind direction signal incoming storms.
- **Lightning:** get off summits, ridges and open water; stay away from isolated tall trees. Crouch low on insulating material (pack or pad) on the balls of your feet if caught in the open; do not lie flat. In a group, spread out. Wait 30 minutes after the last thunder before resuming.
- Never camp in dry streambeds, gullies or low spots that could flood.

4. Navigation and not getting lost

If you are lost

- Stop. Sit down, drink water, calm down. Think about when you last knew where you were.
- In most cases **stay put**: it is easier for searchers to find a stationary person. Make yourself visible and warm.
- Move only if you know where you are heading, there is no hope of rescue nearby, or the location is unsafe. Mark your direction (blazes, stacked rocks, arrows of sticks) so searchers can follow.
- If you must move: follow streams downhill, which usually lead to roads or settlement, but beware of cliffs, falls and swamps; avoid thin ice; prefer logging roads and trails.

Map and compass

- Topographic maps: contour lines show elevation (closely spaced = steep, widely spaced = gentle); blue = water, green = vegetation, black = man-made features, brown = contours. Always read the legend and scale.
- **Declination:** the difference between true north and magnetic north varies by place; adjust for it, and check the current figure on your map.
- To take a bearing: point the direction-of-travel arrow at the target, rotate the dial until the needle sits in the orienting arrow, and read the bearing. To follow it, hold the compass level and walk to landmarks along the line.
- A back bearing is the reverse (add or subtract 180) and helps you check your position or return.
- Orient the map by aligning its north with the compass needle (after adjusting for declination).
- Keep compasses away from metal, vehicles, cell phones and power lines because they interfere with the needle.
- GPS is helpful but batteries fail and signals drop. Treat it as a backup to map and compass.

Without a compass

- **Shadow-tip method (FM):** plant a straight 1 m stick upright, mark the tip of the shadow, wait 10 to 15 minutes, mark again. A line through the two marks runs roughly west to east (the first mark is west). Standing with the first mark on your left gives north ahead.

- **Watch method:** in the northern hemisphere, point the hour hand at the sun; south lies halfway between the hour hand and 12 o'clock (use standard time).
- **Stars:** in the northern hemisphere, find the Big Dipper and extend the line through its two front stars about five times their separation to reach Polaris (due north). Cassiopeia is on the opposite side of Polaris. In the southern hemisphere use the Southern Cross.
- **Moon and sun:** the sun rises in the east and sets in the west (roughly). Do not rely on moss growing on trees; it is unreliable.
- **Finger method (daylight left):** hold your hand at arm's length with fingers horizontal between the sun and the horizon; each finger width is about 15 minutes of daylight.
- **Improvised compass:** magnetize a needle by stroking it in one direction on silk or hair, then float it on a leaf in still water.

Pace counting (Survivor Library)

- A typical pace is about 75 cm. Count 66 double paces as one tally, about 99 m. Ten tallies are about 1 km.
- Keep a tally on a string or by moving pebbles between pockets.

5. Water

How much

- Drink at least 2 to 3 liters per day, even in cold weather. In hot climates and heavy work, needs can exceed 10 liters; in desert heat the FM estimates up to about 19 L per day.
- A 2 percent loss of body water impairs performance; around 5 percent causes severe symptoms; 10 percent or more is dangerous.
- Drink regularly in small amounts. Do not ration water and then sit with a full canteen; drink it and look for more.
- Dark urine is a warning sign. Alcohol and caffeine do not help.
- Do not eat snow or ice straight (it lowers body temperature); melt it first, preferably with stove or fire. Do not drink urine or seawater.
- If water is scarce, avoid eating (digestion uses water), especially protein.

Finding water

- Follow valleys downhill, game trails, and flocks of birds. Look at the bottom of dry stream bends, the outside of bends, and green vegetation.
- Dew: wipe it from grass and leaves with cloth in the early morning and wring it out.
- Rainwater: collect with a tarp or poncho. Sap from certain trees: do not keep for more than 24 hours.
- **Solar still:** dig a hole about 1 m wide and 60 cm deep, place a container in the center, cover with clear plastic held down by soil, and set a small stone in the middle of the plastic over the container. Expect only about 0.5 to 1 liter per 24 hours, so build several (about three per person).

- Aboveground bag still: tie a clear bag over green, non-poisonous leafy branches and let condensation collect.

Purification

Treat all surface water, even if it looks clear. Water-borne diseases include giardia, cryptosporidium, dysentery, cholera, typhoid and others.

- **Boil:** a rolling boil for at least 1 minute at sea level, plus 1 minute per 300 m of altitude; or boil 10 minutes anywhere to be safe. Survivor Library suggests at least 3 minutes for stagnant water.
- **Iodine (2 percent tincture):** 5 drops per liter or quart of clear water, 10 drops if cloudy or cold; wait 30 minutes. (Survivor Library gives 9 drops per quart.)
- **Tablets:** follow the label. Chlorine dioxide also kills cryptosporidium.
- **Filters:** a good commercial filter (0.2 micron or smaller) removes protozoa and bacteria but not necessarily viruses; combine with chemicals if in doubt.
- **Improvised filtering** (cloth, sand, charcoal, standing water for 12 hours) clears sediment and odor only. It does not make water safe. Always boil or treat afterwards.
- UV pens and distillation are also valid methods (Marshall). Bleach (unscented, 5 to 9 percent) can be used at about 2 drops per liter if nothing else is available.

Water storage at home or in camp (Marshall/FEMA)

- Plan 0.5 to 1 gallon (2 to 4 L) per person per day. That is about 3 gallons for 3 days, 21 gallons for 3 weeks and 90 gallons for 3 months, per person. One gallon weighs about 8 lb (3.6 kg).
- Use food-grade containers, keep them cool and dark, and rotate stock.

6. Shelter

Choosing a site

- Look for shelter from wind and weather, level ground, nearby wood and (not too close) water, and visibility to rescuers.
- Camp partway up a hill rather than the valley floor (cold air pools there) or the exposed summit.
- Check overhead: avoid dead branches (widow-makers), loose rock, and dead standing trees. Avoid animal trails, insect nests, flood plains, and spots beside a loud brook (it masks sounds and may rise).
- Do not make shelter larger than needed: smaller is warmer. Insulate yourself from the ground; never sleep directly on cold ground or snow.
- Build it before dark and before you are exhausted. Never use a fire inside a closed shelter without ventilation: carbon monoxide kills.

Debris hut

- Make a ridgepole about 2.5 m long (a bit taller than you), prop one end on a stump or forked tree about knee to waist height, and put the other on the ground.

- Lay ribbing (sticks) along both sides of the pole, then cover with brush or netting, then pile at least 1 m of leaves, grass or pine boughs. Add a 30 cm inside layer of dry bedding. Plug the door with a bundle of debris.
- This is very warm because the thick insulation traps body heat.

Lean-to

- Place a horizontal pole between two trees about 2 m apart (or lash to two forked uprights) and lean 5 to 8 poles about 3 m long against it at roughly 45 degrees.
- Weave brush through the poles; shingle boughs from the bottom up so water runs off; thick bedding on the floor.
- Build a fire reflector wall of logs on the open side to bounce heat in, and keep the fire a safe distance away. Add a drying rack.

Other options

- **Poncho or tarp:** a lean-to, tent, or A-frame; the quickest shelter.
- **Natural shelters:** caves, rock overhangs and fallen trees. Check for snakes, ticks, bats and loose rock. A pit under a fallen tree can be roofed with boughs. Survivor Library suggests marking a cave entrance with string so you can find your way out.
- **Wigwam or tepee:** poles leaned together with a covering of bark or boughs.
- **Swamp bed:** raise a platform above wet ground with logs and cover with boughs.
- **Tree-pit shelter (snow):** dig around the base of a large evergreen, use lower branches as a roof, and line with boughs.
- **Snow cave:** dig into a deep drift (about 3 m deep) with walls about 30 cm thick. Raise the sleeping platform above the entrance so warm air stays in. Provide at least two vent holes (Survivor Library) and keep a tool inside in case the entrance blocks. Mark the outside.
- **Snow trench:** dig a trench, roof with branches and snow blocks.
- **Desert:** a trench about 45 to 60 cm deep can be 16 to 22 C cooler. Use layered cloth cover with an air gap 30 to 45 cm above for shade; stay off the hot ground.
- **Beach:** a shade shelter from driftwood and clothing above the high tide line.

Sleeping warm

- Insulate under you with boughs, dry grass (at least about 6 mm / 1/4 inch thick wraps), leaves, or pack.
- Heat stones in the fire, wrap them in cloth or place them under the bedding (not directly touching skin), and replace as needed.
- A contractor bag or space blanket makes an effective emergency bag. Reflective foil placed behind you throws back heat.
- Eat a sweet snack before bed, cover your head, loosen tight boots and clothing, and keep clothing dry.

7. Fire

Principles

- Fire needs heat, fuel and oxygen. Clear a circle about 1 m wide down to bare soil, away from overhanging branches and roots.
- Do not build on wet or porous rocks (they can explode). On snow, build on a base of green logs. A Dakota hole (a pit with an air tunnel) shields wind and hides the fire.
- Gather three grades before you light: **tinder** (dry grass, birch bark, dryer lint, cotton balls with petroleum jelly, fungi, char cloth, pine needles, fuzz sticks), **kindling** (pencil-thin to thumb-thick dry sticks) and **fuel** (larger wood).
- Firewood quantity rule (Maine Woods): estimate what you will burn in one hour, multiply by the number of dark hours, then double it. Keep it dry and covered.
- Standing dead wood is drier than wood on the ground. Split logs to reach dry interiors.

Fire lays

- **Tepee:** kindling leaned in a cone over tinder; fast and hot.
- **Lean-to:** kindling leaned against a log over tinder.
- **Cross-ditch / pyramid:** layers of wood in a stack; burns long and makes coals for cooking.
- Light on the upwind side so the wind carries flame into the fuel.

Ways to light

- Waterproof matches or lighter (carry several), ferro rod (metal match), magnifying lens, battery and steel wool, flint and carbon steel, and friction methods (bow drill, fire plow) as last resorts. For friction use dry, non-resinous, softwood from standing dead wood, not wood lying on wet ground.
- Waterproof matches by dipping heads in wax or nail polish, or store in a sealed container.
- **Fire in the rain:** look under dense evergreens, split wet wood, shave dry inner wood into fuzz sticks, use resinous wood or commercial fire starters, and keep the fire sheltered by a tarp or a log reflector.

Fire safety

- Keep water or soil nearby. Never leave a fire unattended.
- Bank coals with ash to keep them alive overnight.
- Make sure the fire is completely out: drown, stir, and feel the ashes with your hand. Check local fire bans.
- Dry clothes in stages near the fire, not too close; do not melt synthetics.

Home-disaster fire materials (Marshall)

Fire starters to store: cotton balls with petroleum jelly, dryer lint, newspaper, tree fungus, dry moss, char cloth, small kindling, charcoal, dry leaves or needles, and bark. Fire provides warmth, water purification, cooking, light, signaling and morale.

8. Food

- Food is a lower priority than shelter and water; a person can go a week or more with little food. About 500 kcal per day can keep you alive for some time but not working hard; normal need is about 2,000 kcal or more.
- Only eat plants you can identify with certainty. If unsure, do not eat it.
- Warning signs of danger: milky sap, beans or bulbs inside pods, almond scent, spines, shiny leaves, three-leaf pattern (poison ivy/oak), umbrella-shaped flower clusters (carrot family look-alikes such as water hemlock). Avoid red and white berries (baneberry and many others) and anything that looks like bean or melon plants (Survivor Library).
- **Universal Edibility Test (FM):** only for desperate situations and only if you can eat nothing else for the entire test. Test one plant part at a time: smell it, touch it to skin, then lips, tongue, chew and hold for 15 minutes, then swallow a small amount and wait 8 hours before eating more. This process takes about a full day per plant and is not a guarantee. Some deadly plants (mushrooms) do not pass reliably. Avoid mushrooms entirely.
- Fish, insects (cooked), snails and small animals are safe sources when thoroughly cooked. Cook everything to kill parasites. Do not eat anything that appears sick.
- Food storage (Marshall): stock shelf-stable foods you eat normally, such as canned goods, rice, beans, pasta, oats, peanut butter, dried fruit and energy bars; store in a cool dry place and rotate. Have a manual can opener.

9. Signaling and rescue

- Signals work in groups of three: three fires (in a triangle or line, about 25 to 30 m apart), three whistle blasts, three shots, three flashes. Keep fires ready to light.
- **Smoke:** add green leaves, moss or wet material for white smoke (best against dark backgrounds); rubber, oil or plastic for black smoke (best against snow or light backgrounds).
- **Mirror:** sweep the reflected light toward aircraft or distant people. Do not flash rapidly or into a cockpit directly; sweep across the horizon.
- **Whistle:** carries farther than your voice (up to about 1.6 km) and takes less energy.
- **Ground signals:** large contrasting shapes at least 1 m wide by 6 m long; an X, SOS, or an arrow. Make a big X with legs about 4.5 m (15 ft) in snow or on open ground. Stamp or dig in snow and fill with boughs.
- Bright clothing (blaze orange), reflective emergency blankets and a tarp laid in a clearing are visible from the air.
- Radio/phone: if you have coverage, call early; text can work when voice does not. Conserve the battery.
- When rescuers arrive or you hear an aircraft, move to an open area and make yourself as large and visible as possible. Stay with your signals.

10. First aid and environmental injuries

Basic rule: check the scene for safety, then check airway, breathing and circulation. Call for help early.

Bleeding

- Apply direct pressure with a clean cloth for at least 10 to 15 minutes; add more cloth rather than removing it. Elevate the limb above the heart.
- If it still bleeds after about 30 minutes, apply a pressure dressing. A tourniquet is a last resort for life-threatening limb bleeding: place it 5 to 10 cm above the wound, tighten until bleeding stops, note the time, and do not loosen.

Shock

- Signs: pale cool clammy skin, rapid pulse, weakness, confusion. Lay the person down, elevate the legs (unless injured), keep warm, give nothing by mouth unless fully alert, and treat the cause.

Rescue breathing / CPR

- If not breathing, call for help and start CPR as you have been trained: chest compressions (about 100 to 120 per minute) and rescue breaths. Older guides describe rescue breathing at 12 to 15 breaths per minute for adults and 20 to 25 for children; get current training.
- Heart attack: sit the person comfortably, call for emergency help, and give aspirin if not allergic and the person is conscious.

Fractures, sprains and dislocations

- Immobilize the joint above and below the break with a splint padded with clothing. Check circulation beyond the splint.
- Sprains: RICE (Rest, Ice, Compression, Elevation). Do not try to force a dislocated joint back unless trained; splint and evacuate. Survivor Library's advice to give aspirin for dislocations is dated.
- Head injury or concussion: watch for confusion, vomiting, drowsiness, unequal pupils; keep the person still and get help.

Wounds, burns and blisters

- Clean wounds with clean (treated) water, cover with a sterile dressing, and watch for infection (redness, swelling, pus, fever).
- Burns: cool with running water for 15 to 20 minutes, cover loosely with clean non-stick dressing; do not pop blisters or apply butter.
- Blisters: protect with padding; drain only if needed with a clean needle, keep covered. Stop and treat hot spots early with tape.

Hypothermia

- Signs: shivering, slurred speech, stumbling, confusion; severe cases stop shivering and become drowsy.
- Treat: get out of wet clothing and wind, insulate from the ground, rewarm the trunk first (warm packs on chest, armpits and groin), give warm sweet drinks to a conscious person, and rewarm gradually. Handle gently.
- Watch for an **afterdrop** and rebound when cold blood returns from the limbs. Do not give alcohol.

Frostbite, trench foot

- Frostbite: skin turns white or waxy, numb. Do not rub. Rewarm in warm (not hot) water about 37 to 39 C for 20 to 30 minutes, only if there is no risk of refreezing; do not thaw if you must keep walking on it.
- Trench or immersion foot: keep feet dry, change socks, and air them often; treat by drying and warming gently and elevating.

Heat illness

- Heat cramps: rest in shade, drink water with a pinch of salt (about 0.25 tsp per liter).
- Heat exhaustion: heavy sweat, headache, dizziness, nausea. Rest in shade, cool the body, sip fluids.
- Heat stroke (emergency): hot dry skin or confusion, high temperature. Cool the person immediately (water, fanning, ice packs at neck, armpits and groin) and get urgent help.
- Prevent: wear light loose clothing and a hat, avoid heavy work at midday, drink before you are thirsty.

Bites, stings and ticks

- **Snakebite:** keep calm and still, keep the limb at about heart level, remove rings, and get medical care urgently. Do not cut, suck or use a tourniquet.
- Bee/wasp stings: scrape off the stinger, wash, apply cold. Watch for allergic reaction (difficulty breathing, swelling); use epinephrine if prescribed.
- **Ticks:** use permethrin-treated clothing and DEET on skin; tuck pants into socks; check yourself daily. To remove, grasp as close to the skin as possible with fine tweezers and pull steadily straight out, parallel to the skin. Do not use petroleum jelly or heat (the FM's advice is dated). Watch for a bull's-eye rash and fever; see a doctor.
- Insects: use repellent, head net, and long sleeves; avoid camping near still water.

Other cold and sun problems

- **Snow blindness:** wear sunglasses or improvised slit goggles from bark, cardboard or cloth. Treat with rest in the dark and cool compresses.
- **Sunburn:** use sunscreen, cover up, and remember that snow and water reflect sunlight.
- Diarrhea: replace fluids. Dehydration is the main danger. Treat water and wash hands.
- Headaches are often dehydration; drink and rest.

Cold-water immersion (Maine Woods)

- The first 10 seconds bring the cold shock and gasp reflex; control your breathing before anything else. Then you have about 10 minutes of useful movement to self-rescue, and about an hour before unconsciousness.
- Wear a life jacket, keep clothes on, and get out of the water as fast as possible. Float on your back; assume the HELP position (knees to chest) to preserve heat.

First aid kit (Marshall and others)

Adhesive bandages, sterile gauze, medical tape, elastic wrap, triangle bandage, antiseptic wipes, antibiotic ointment, gloves, tweezers, scissors, safety pins, pain relievers, antihistamine, burn gel, moleskin, thermometer, emergency blanket, CPR face shield, and any personal medications.

11. Special environments

Desert (FM)

- Seven factors: low rainfall, intense sunlight and heat, wide temperature range, sparse life, rocky and sandy terrain, high mirage and glare, and flash floods.
- Drink about 0.5 L per hour below 38 C and about 1 L per hour above. Do not ration water.
- Rest in shade during the day, travel at night or dawn and dusk, stay off the hot ground, and wear loose, full-coverage clothing and a hat.
- Distances are underestimated up to three times. Do not camp in dry wadis (flash floods).
- Shake out boots and bedding for scorpions and insects; wear gloves for handling rocks.
- Tell someone your route and return time; check urine color.

Cold and snow (FM)

- Keep dry and avoid sweating; vent clothing instead of overheating. Eat often and keep hydrated (urine color on snow helps check).
- Use sunglasses or slit goggles; protect face from frostbite; avoid constipation by eating fiber and not holding it in.
- Check ice before crossing; avoid sleeping directly on snow; tend the fire; make a shelter early in the afternoon.

12. Camp craft and safety quick checklist

- Tell someone; carry map, compass, fire, whistle, knife, light, water, food, and first aid.
- Wear layers; carry extra; stay dry.
- Make camp before dark; site safe from overhead hazards and floods; keep wood supply ample.
- Keep a clean camp; store food away from tents and out of reach of animals (hang or use a bear canister where required). Do not cook near sleeping areas.
- Treat water. Drink steadily.
- If lost: STOP; stay put; build shelter, fire and signals; use the rule of threes.
- Know your limits: turn around early, and do not hike alone in poor weather or in unfamiliar terrain.
- Boat safety (Maine Woods): wear a life jacket, file a float plan, check weather, carry a whistle and bail, and do not overload. Kids: attach whistles and bright clothing, teach them to hug a tree and stay put if lost.

Sources

- US Army FM 21-76, Survival (2006 text edition).
- You Alone in the Maine Woods, Michael F. Richard with Brad Rounds and Bryan Courtois (11th edition, 2020).
- Wilderness Survival, Survivor Library.
- Ultimate Survival Skills Guide, Joe Marshall (home-security and defense sections not used).